UNIVERSITAS GADJAH MADA

Digital Image Processing

Fire Detection in Digital Images Using Enhanced Image Processing and Machine Learning Techniques

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Introduction

Fires can start quickly, especially in environments with enclosed or semi-enclosed environments that pose high risks of fire due to factors such as faulty electrical wiring, unattended cooking or simply carelessness in handling flammable materials. On a larger scale, fire hazards in open environments such as fields, plantations, and forests, require strict supervision. This is why early and accurate fire detection is crucial.

Project Objective

Our objective is to develop a reliable fire detection system that can accurately identify fire occurrences while minimizing false positives. By reducing false alarms, the system ensures that unnecessary worries are avoided, allowing responses to be focused on genuine threats.

To solve this problem we developed a digital image processing model using the AESFERM framework.

Acquisition

- Focus on loading images from Kaggle dataset into the system for training.
- The dataset used contains 760 fire and 760 non-fire images where some has fire-like regions.
- It represents real-world scenarios where fire might occur in environment with various lighting and weather conditions.

Inserted image:

01_Original_Image

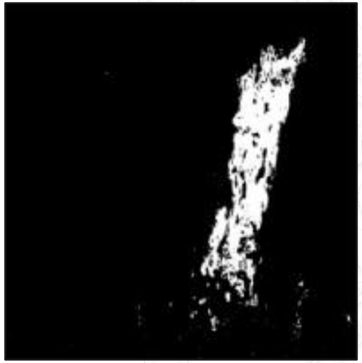
Enhancement

- Color space conversion
  - Separating image into 3 component, H, S, and V
    - Hue helps differentiates between colors
    - Saturations measures the intensity of colors
    - Value measures brightness
  - We want to focus on specific color properties like fire-related hues so isolating fire-like colors in different image conditions become easier.

- Contrast stretching
  - Enhances the brightness channel to improve visibility of fire colors, ensuring consistent quality before passing it to segmentation phase
    - Bright color become brighter
    - Dark color (non-fire) become darker

Segmentation

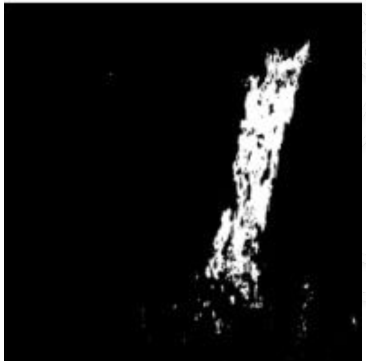
- Color Segmentation
  - isolate fire-like regions in the image based on HSV threshold
    - Hue (10-20): Specifically targets fire-like colors.
    - Saturation (100-255): Fires are vivid, ensuring low-intensity colors are excluded.
    - Value (100-255): Isolates bright regions, eliminating dark or irrelevant pixels.
  - Highlights fire-related pixels effectively, serving as a foundation for further refinement and processing, producing a binary mask output (white for pixels falling within the HSV range)



Segmentation

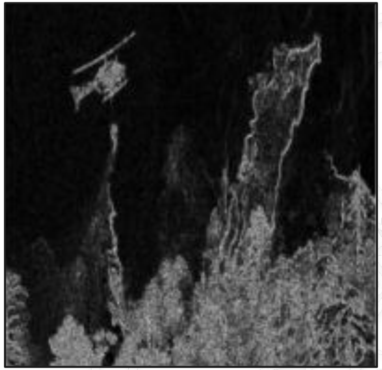
- Otsu's Thresholding
    - Eliminate false positives caused by non-fire objects (e.g., sunsets, red/orange objects) identified during color segmentation so we can focus on isolating regions with bright intensities that resemble fire.
    - **How it works:**
    - Automatically calculates the optimal threshold to separate pixels into two classes:
        - Above the Threshold: Potential fire pixels (white, 255).
        - Below the Threshold: Background/non-fire pixels (black, 0).
    - **How it helps:**
        - Dynamic Adaptation: Automatically adjusts to different lighting conditions, making it suitable for varied images.
        - Accuracy: Effectively distinguishes fire regions from the background based on brightness.
    - Produce a refined mask from the combination of previous fire mask with otsu's binary mask, successfully isolate regions with fire-like characteristics in both color and brightness and reduces false positives

06_Combined_Fire_Mask

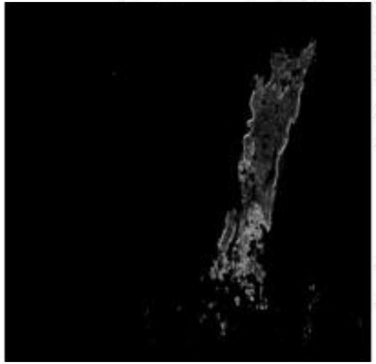


Segmentation

- Sobel Edge Detection
  - Identify the boundary between fire (region of interest) and the background so we can highlight areas with strong intensity changes, which typically correspond to object boundaries.
  - Detect the jagged and irregular contours characteristic of fire
  - **How it works:**
    - Applied to the Value of enhanced HSV, where brightness information is most prominent.
    - Detects changes in x and y axis using 3x3 kernels then combine the gradients to calculate the overall gradient magnitude, which represents the strength of intensity changes.
  - **How it helps:**
    - Highlights edges effectively, aiding in boundary detection.
    - Smoothens the image while preserving essential features, reducing noise.
    - Captures the irregular contours typical of fire, helping differentiate fire from non-fire regions.
  - Produces a refined fire mask that isolates sharp-edges regions, which more likely to be actual fire areas

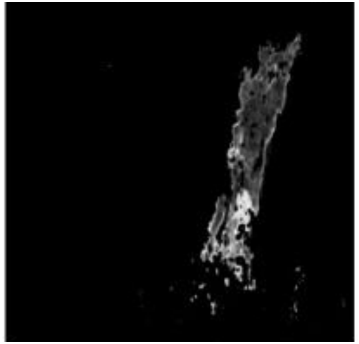


Edges

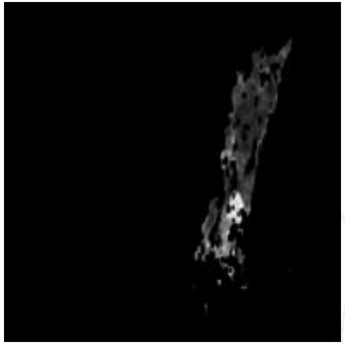


Refined Fire Mask with Edges

Segmentation



Closing

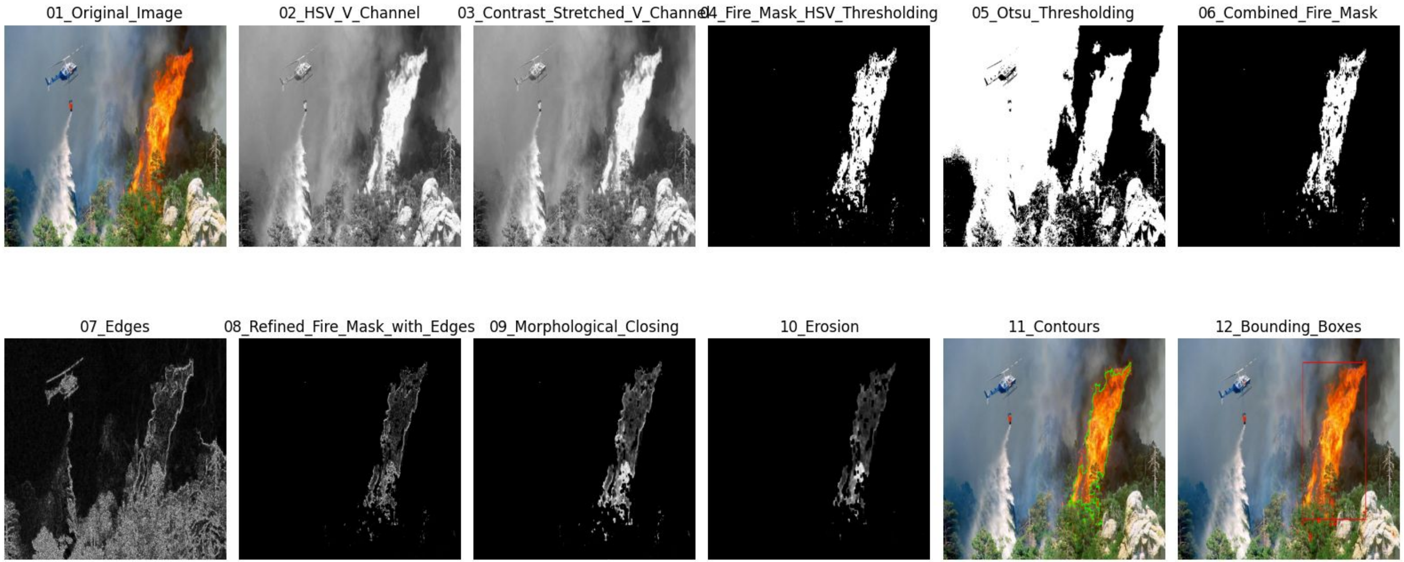


Erosion

- Morphological Operations
  - Address fragmentation and environmental interference in detected fire areas.
  - **Closing**
    - Dilation: Adds white pixels to region edges, connecting separated and fragmented fire regions.
    - Erosion: Removes extra noise created by dilation to smooth the regions.
  - **Additional Erosion**
    - Further cleans the mask by removing isolated noise pixels, ensuring smoother and cleaner region segmentation.
  - Produces a final, noise-free fire mask that isolates fire-like regions accurately.

Segmentation

- Contour Detection
  - Focus on outermost contours, capturing the primary fire areas and ignoring internal details or nested contours.
  - **How it works:**
  - Primary analysis like shape, size, and position of the fire regions are analyzed while ignoring internal fluctuations. Performing feature calculation like:
    - Area: size of fire region
    - Perimeter: indicating the complexity of the fire's shape
    - Aspect ratio: helps determine if region matches typical fire patterns
  - **How it helps:**
    - Filters out irrelevant or noise-based regions like small reflections and Isolated bright spots.
    - Handling overlapping regions by separates overlapping fire regions to avoid treating multiple fires as a single object.
    - Precisely pinpoints fire regions for tracking fire spread, estimating impact


01_Original_Image
02_HSV_V_Channel
03_Contrast_Stretched_V_Channel
04_Fire_Mask_HSV_Thresholding
05_Otsu_Thresholding
06_Combined_Fire_Mask
07_Edges
08_Refined_Fire_Mask_with_Edges
09_Morphological_Closing
10_Erosion
11_Contours
12_Bounding_Boxes

Feature Extraction

- Geometric Features
  - Consist of area, perimeter, aspect ratio, extent, and circularity, used to determine the shape of the fire
  - Helps in distinguishing fire regions from non-fire regions/objects with similar colors but different shapes
- Color Features
  - Done by calculating the mean values of the HSV channels to capture the overall color characteristics of fire

Feature Extraction

- Texture Features using LBP
  - Extract fine-grained texture patterns unique to fire regions.
  - Helps differentiate fire from non-fire regions with similar colors but different textures.
  - **How it works:**
    - Compares pixel intensities in a small neighborhood around each pixel and encodes the comparison as binary patterns, capturing texture details.
    - Focuses on relative intensity differences instead of absolute pixel values
  - **How it helps:**
    - Helps eliminate irrelevant patterns or noise.
    - Enhances performance in Support Vector Machine (SVM) classification later in the process by providing meaningful texture features.

- Gradient-based Features
  - Captures texture and intensity changes within the fire region by calculated the mean and standard deviation of gradient magnitudes using Sobel filters
  - Highlights directional intensity variations, which are characteristic of fire patterns.

Feature Representation

- Feature vector
  - Combines all extracted features (shape, color, edge, gradient, texture) into a single vector for each detected fire region.
  - Ensures all relevant data is consolidated for further analysis or classification.

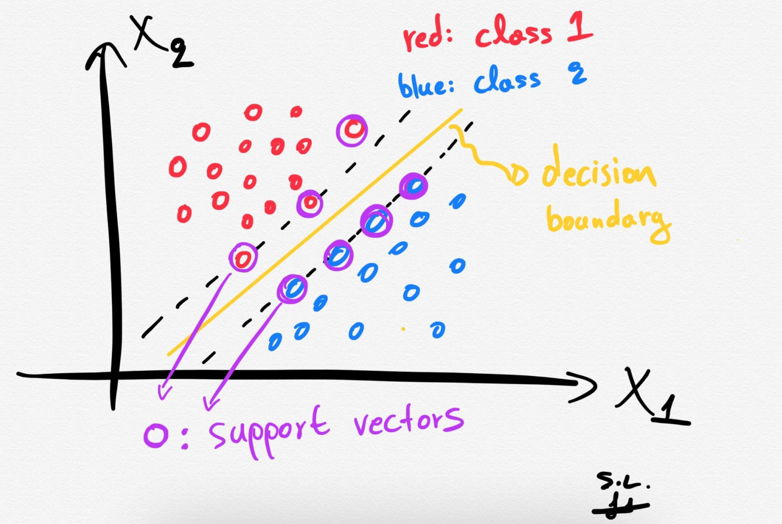
- Normalization
  - Adjusts feature vectors to a consistent scale.
  - Ensures fair contribution of all features during the model's learning process, preventing dominance by features with larger values.

Feature Representation

```python
# Add features to the list
feature_vector = [
    area, perimeter, aspect_ratio, extent, circularity,
    *mean_hsv, edge_density, grad_x_mean, grad_x_std, grad_y_mean, grad_y_std,
    lbp_mean, lbp_std
]
```

Area	Perimeter	Aspect_Ratio	Extent	Circularity	Mean_H	Mean_S	Mean_V	Edge_Density	Grad_X_Mean	Grad_X_Std	Grad_Y_Mean	Grad_Y_Std	LBP_Mean	LBP_Std
273.0	100.627416	1.291667	0.366935	0.338797	14.417445	186.426791	177.146417	0.468853	-6.29595	175.093147	-29.498442	233.287038	4.788162	3.011709
42048.5	1122.208151	1.0	0.672776	0.419579	13.68152	208.292536	190.213157	0.162898	0.394431	49.925621	-0.796422	35.837616	5.017303	2.324174
1340.5	260.325901	0.653333	0.364762	0.248566	13.371585	198.202186	247.876366	0.155001	-0.657104	38.0667	-0.244536	34.356103	4.890027	2.601399
607.0	128.627417	1.029412	0.510084	0.461032	14.64275	198.553064	206.122571	0.42078	19.763827	160.825071	-1.4858	91.723696	4.639761	2.772431
9164.0	771.906635	1.254386	0.56214	0.19327	15.608563	243.55882	228.915626	0.253334	-2.058348	116.090057	-2.27558	114.483495	4.701543	2.79527

Feature Matching



Support Vector Machine (SVM) chosen for its ability to learn patterns from extracted features and make prediction on new inputs. It detect fire in images by comparing known fire characteristics with features extracted from the input image.

- Effective for High-Dimensional Data
- Reliable Performance with Small and Moderate Datasets

Divided into two phases:

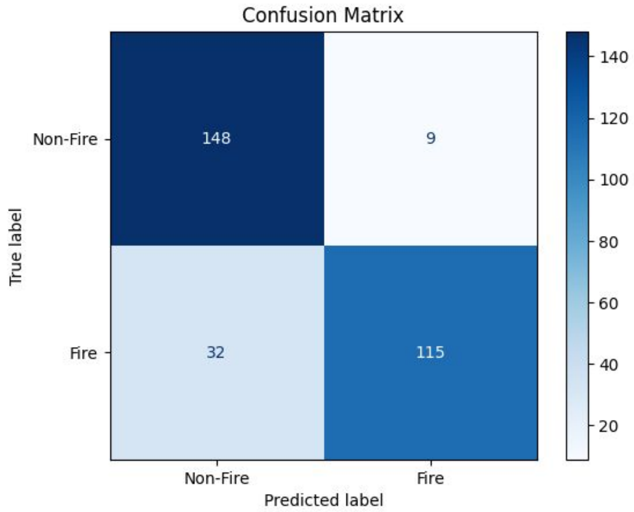
- **Training Phase:** SVM learns the relationship between feature vectors and their corresponding labels (fire/non-fire).
- **Testing Phase:** SVM uses feature vectors from new images to classify regions as "fire" or "non-fire."

Result

Class	Precision	Recall	F1-Score	Support
0 (Non-Fire)	0.84	0.97	0.90	157
1 (Fire)	0.97	0.80	0.88	147
Accuracy	-	-	0.89	304
Macro Average	0.90	0.89	0.89	304
Weighted Average	0.90	0.89	0.89	304

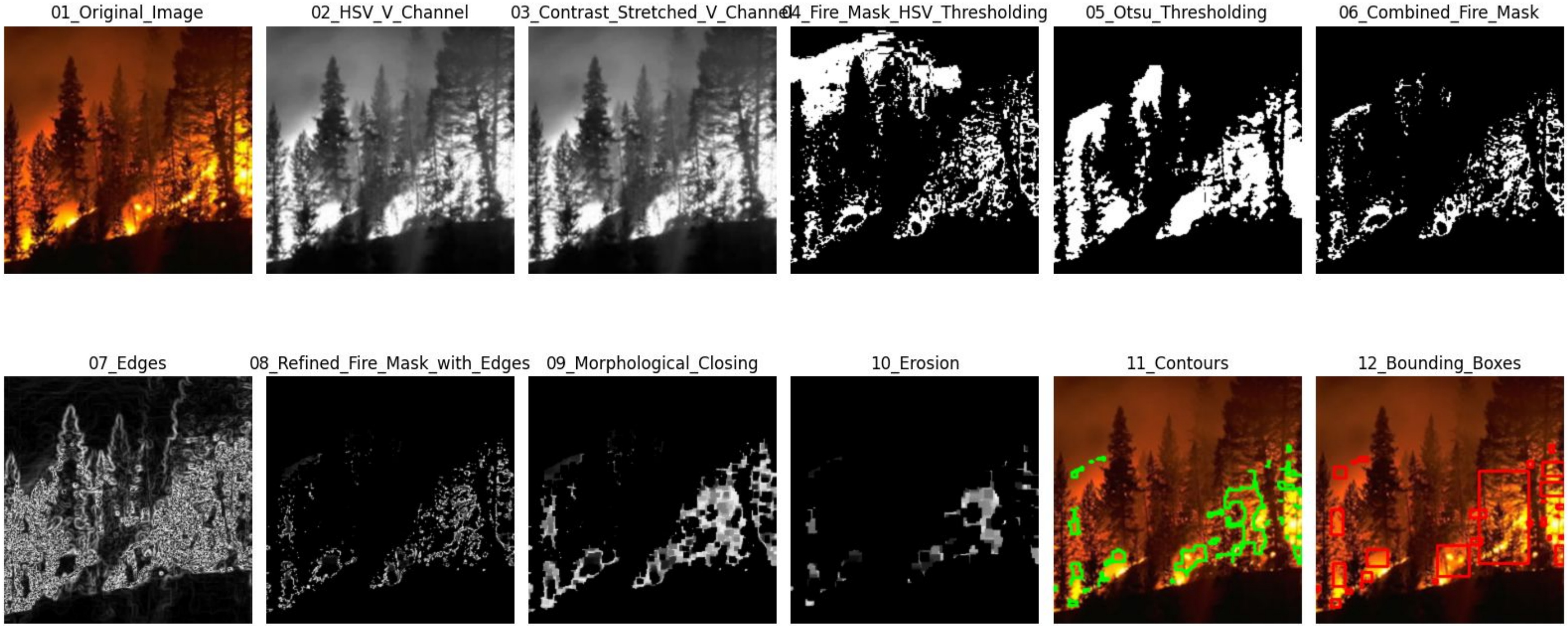
TABLE I

RESULT OF CLASSIFICATION RESULTS WITH ACCURACY OF 89.14%



Result


No Fire Detected
No Fire Detected
No Fire Detected
No Fire Detected
No Fire Detected



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07_Edges
08_Refined_Fire_Mask_with_Edges
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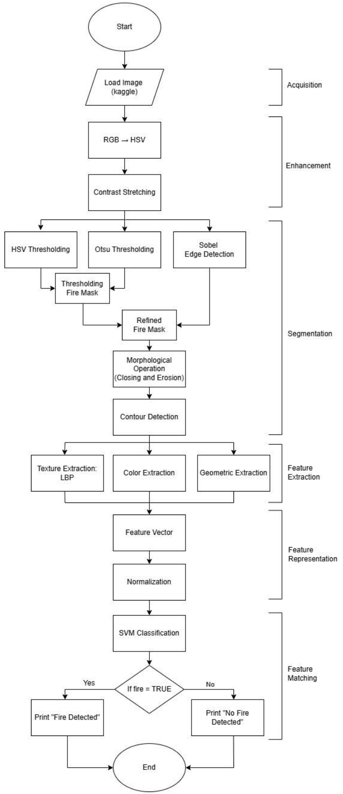
10_Erosion
11_Contours
12_Bounding_Boxes

Result


Fire Detected
Fire Detected
No Fire Detected
No Fire Detected
Fire Detected
Fire Detected
Fire Detected
No Fire Detected
Fire Detected
No Fire Detected

Pipeline



Start
Load Image (kaggle)
Acquisition
RGB → HSV
Contrast Stretching
Enhancement
HSV Thresholding
Otsu Thresholding
Sobel Edge Detection
Thresholding Fire Mask
Refined Fire Mask
Morphological Operation (Closing and Erosion)
Contour Detection
Segmentation
Texture Extraction: LBP
Color Extraction
Geometric Extraction
Feature Extraction
Feature Vector
Normalization
Feature Representation
SVM Classification
If fire = TRUE
Yes
No
Print "Fire Detected"
Print "No Fire Detected"
Feature Matching
End

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Demo

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Thank You